Sara Abouroumia

M.S. Student in Computer Engineering, Istanbul Technical University
Istanbul, Türkiye · abouroumia26@itu.edu.tr · linkedin.com/in/sara-abouroumia-7b5665210

Research Interests

Using artificial intelligence as a tool to address cybersecurity threats, including AI-driven network security, explainable and robust intrusion detection, 5G mobile network security, and the security of AI-generated code and agentic systems.

Education

Istanbul Technical University (İTÜ) — Istanbul, Türkiye
M.S. in Computer Engineering (thesis track) — *2026 – present*

- Supervisor: Prof. Dr. Abdul Halim Zaim. Thesis focus: using AI to address cybersecurity threats, with an emphasis on intrusion detection and network security.
- Current courses: Deep Learning, Machine Learning, Computer Networks, Special Topics in Computer Engineering.

Abdullah Gül University (AGÜ) — Kayseri, Türkiye
B.S. in Computer Engineering, with honors (instruction in English) — *2019 – 2024*

- GPA: 3.45 / 4.00. Final-year term GPAs of 3.80 and 4.00.
- Relevant coursework: Network Security (A), Computer Networks (A−), Machine Learning in Python (A−), Probability and Statistics (A), Linear Algebra (A), Operating Systems (A), Computer Organization (A).
- Capstone (A): *Diabetic Foot Ulcer Classification Using CNN Transfer Learning and TPU Acceleration*.

Research Experience

Istanbul Technical University — Istanbul, Türkiye
Graduate Researcher (unfunded), MRGuardAI project — *Oct 2026 – present*

- Advisor: Doç. Dr. Mehmet Ali Ertürk. Research on managed-runtime security for LLM agents, detecting unsafe agent behavior by instrumenting the JVM and .NET runtimes.
- Reviewed recent work on agent security, from tool-call guards to kernel-level monitors (eBPF), and narrowed the project's novelty to the managed-runtime layer between them.
- Drafted the 12-month research plan, with reference stacks on Spring AI (JDK 25) and Microsoft Agent Framework (.NET 10) and target venues including IEEE ICC 2027 workshops and IEEE PIMRC 2027.
- Now preparing agent behavior datasets for the first milestone, a JVM-based detection prototype.

Professional Experience

Nextarp B.V. — Netherlands (remote)
Full Stack Software Engineer, Cloud and DevOps — *Jul 2024 – Oct 2026*

- Architected, deployed and maintained full-stack web and mobile applications (Spring Boot, React, React Native) serving production traffic for training management and ERP systems, owning each system from design to production.
- Built an agentic development lifecycle (ADLC) system of automated bots that handle code review, testing and a multi-stage security gate covering static analysis (SAST), secrets scanning and dependency vulnerability checks (SCA), authenticating to GCP without stored keys through Workload Identity Federation.
- Engineered Spring Boot backend services and RESTful APIs integrating MySQL, Redis caching and Stripe payments, using efficient data access patterns for low latency under real load.
- Set up CI/CD pipelines with Docker and GitHub Actions for zero-downtime deployments to GCP and Azure, and managed releases through Microsoft Partner Center, Google Play and the App Store.
- Extended ERPNext/Frappe with custom modules and REST integrations, and developed Microsoft Word add-ins on .NET, onboarding quickly onto an unfamiliar stack.

Outlier — Remote
AI Model Evaluator (freelance) — *2025*

- Evaluated and wrote training data for large language models, including function-calling and code reasoning tasks; designed an original competitive programming problem for an evaluation assessment.

MAZAKA Kayseri, Türkiye
React Developer (part-time), previously Web Development Intern *Jun 2022 – Dec 2023*

- Developed React web applications; as an intern, built a React and Node.js application with JWT-based authentication.

Selected Projects

Security Gates for AI-Written Code Aug 2026
micro1 Frontier Engineering Challenge

- Built a repository-onboarding agent and merge gates that check AI-generated code for security flaws, malicious content, insecure dependencies and regressions, ending in a sign-off report for a human reviewer. Evaluated on Python and TypeScript test repositories.

Explainable AI for Network Intrusion Detection Fall 2026
İTÜ Computer Networks course, paired paper project (in progress)

- Studying explainability methods for traffic classification and intrusion detection models.

Medical Image Classification with CNNs 2023
AGÜ capstone project

- Classified diabetic foot ulcer images with CNN transfer learning, trained using TPU acceleration.

Technical Skills

ML and data	Python, scikit-learn, CNNs and transfer learning, SQL
Security and cloud	SAST, SCA, secrets scanning, Workload Identity Federation, Docker, GitHub Actions, GCP, Azure
Software	Java (Spring Boot), C# (.NET), TypeScript, React, React Native, Next.js, Node.js, MySQL, Redis

Leadership

Co-founder, Kayseri Social Run Kayseri, Türkiye

- Co-founded a youth running community of more than 300 members and built its web platform (Next.js, Postgres, Vercel).

Languages

Arabic (native), English (fluent), Turkish (basic, conversational)

References

Prof. Dr. Abdul Halim Zaim
Istanbul Technical University
M.S. thesis supervisor
azaim@itu.edu.tr

Prof. Dr. Zafer Aydın
Abdullah Gül University
Head of Computer Engineering
zafer.aydin@agu.edu.tr

Asst. Prof. Abdulkadir Köse
Abdullah Gül University
Computer Engineering
abdulkadir.kose@agu.edu.tr